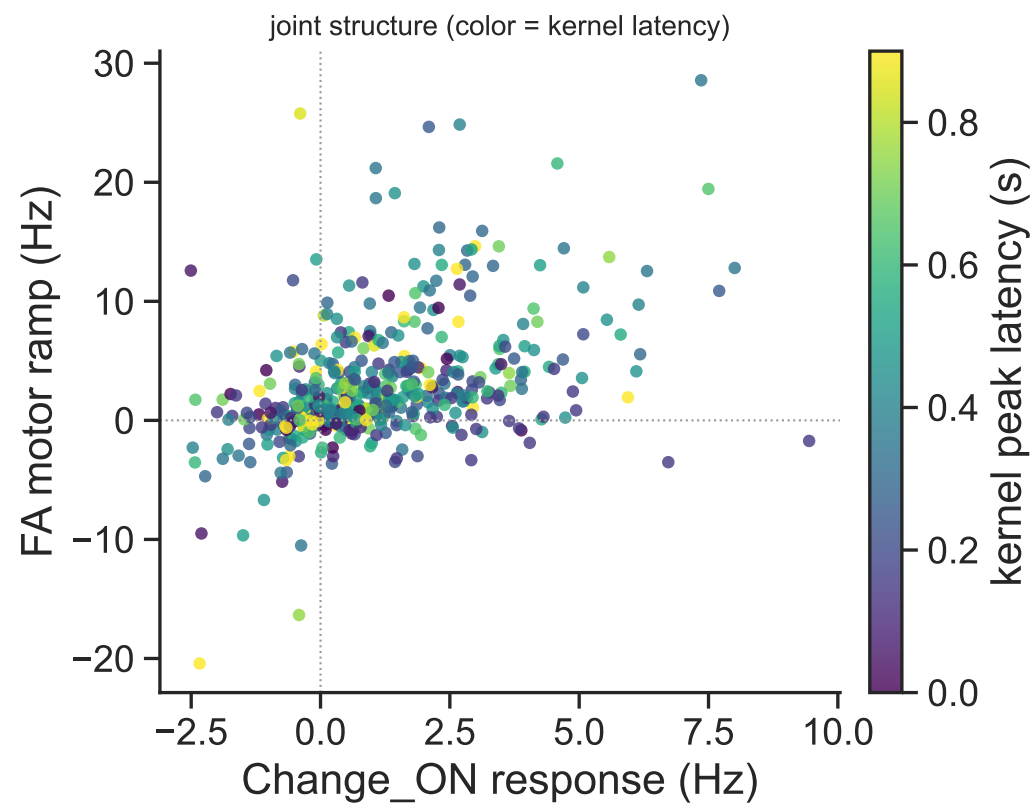
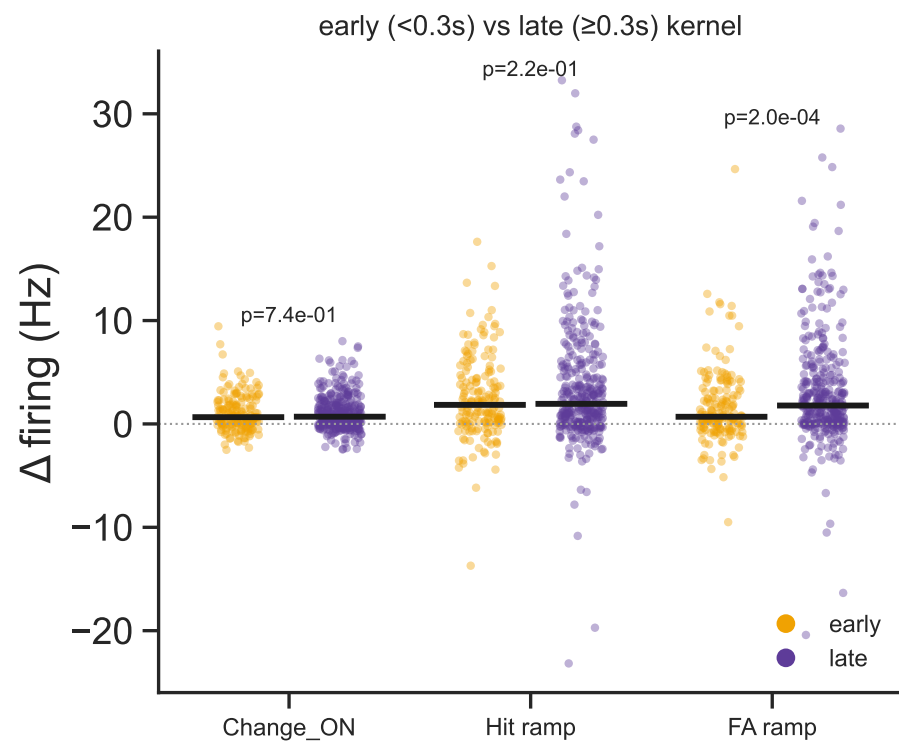
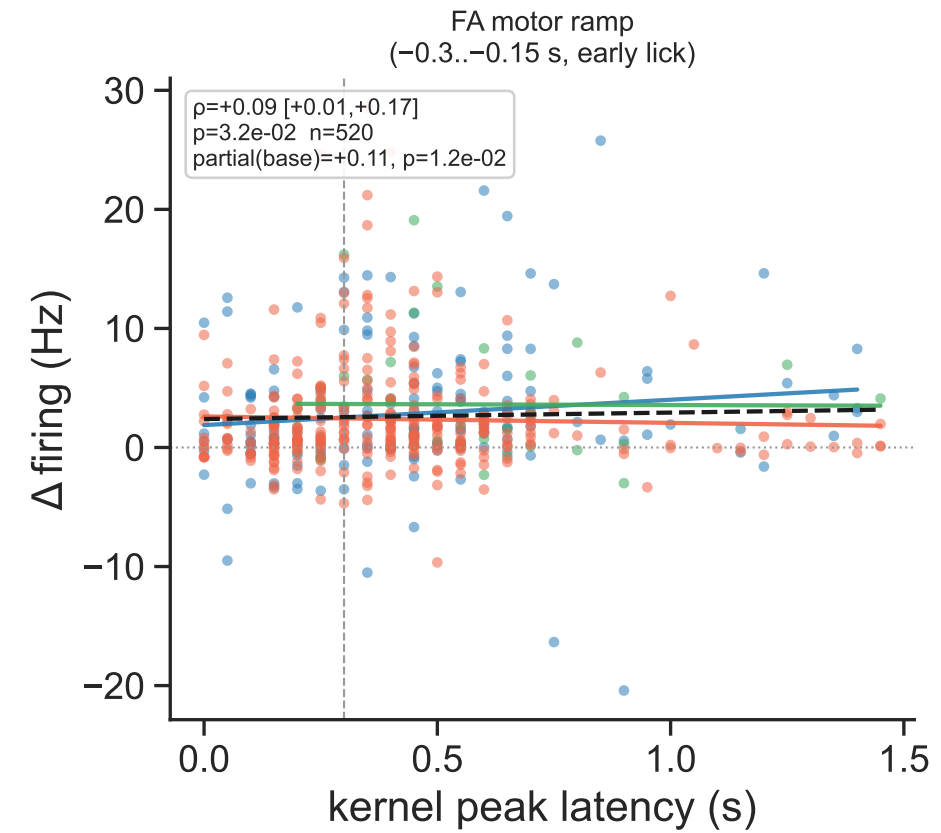
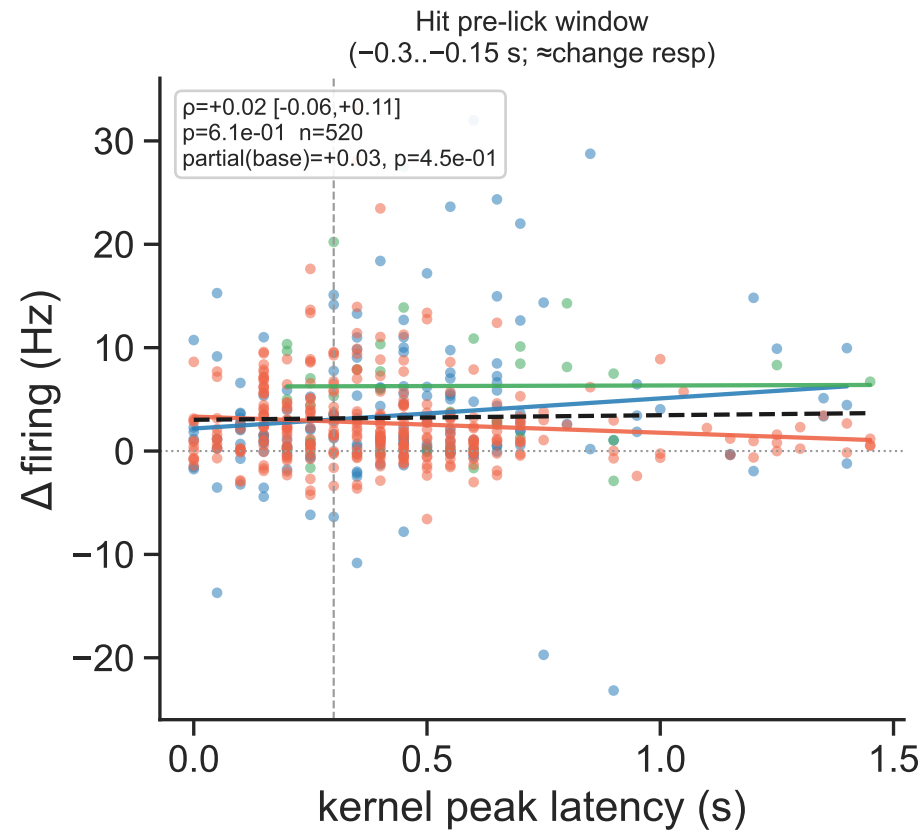
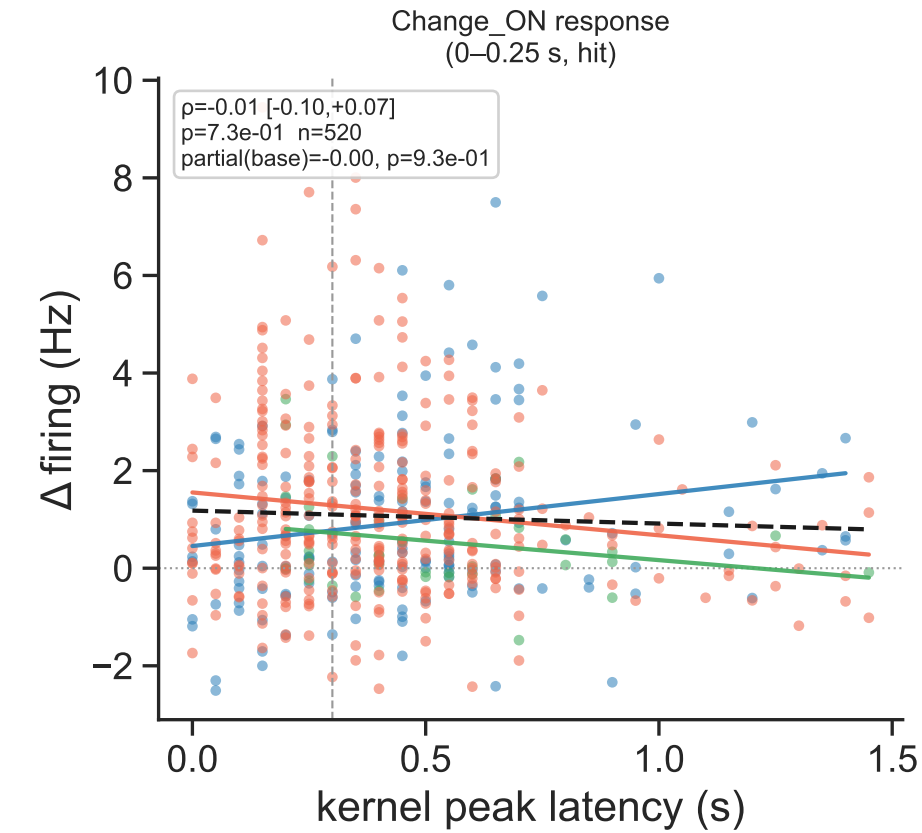


TF-kernel latency is largely DECOUPLED from outcome-response magnitude across the population
 null when pooled ($\rho \approx 0$, dashed line); a weak POSITIVE relation in BG_046 DMS only, flat/reversed in BG_031 VMS — coloured lines = per-mouse fits



Spearman ρ (kernel latency, response)

```
[change_on] pooled  $\rho = -0.015$   $p = 7.34e-01$ 
partial =  $-0.004$  ( $p = 9.25e-01$ ) | BG_046
 $\rho = +0.20$  ( $p = 9e-03$ ,  $n = 162$ ) · BG_039
 $\rho = -0.17$  ( $p = 3e-01$ ,  $n = 39$ ) · BG_031
 $\rho = -0.10$  ( $p = 9e-02$ ,  $n = 319$ )
[hit_ramp] pooled  $\rho = +0.022$   $p = 6.11e-01$ 
partial =  $+0.034$  ( $p = 4.45e-01$ ) | BG_046
 $\rho = +0.18$  ( $p = 2e-02$ ,  $n = 162$ ) · BG_039
 $\rho = +0.05$  ( $p = 8e-01$ ,  $n = 39$ ) · BG_031
 $\rho = -0.11$  ( $p = 5e-02$ ,  $n = 319$ )
[fa_ramp] pooled  $\rho = +0.094$   $p = 3.19e-02$ 
partial =  $+0.111$  ( $p = 1.17e-02$ ) | BG_046
 $\rho = +0.20$  ( $p = 1e-02$ ,  $n = 162$ ) · BG_039
 $\rho = +0.05$  ( $p = 8e-01$ ,  $n = 39$ ) · BG_031
 $\rho = +0.03$  ( $p = 6e-01$ ,  $n = 319$ )
[change_on] early ( $n = 174$ ) med =  $+0.66$  vs late ( $n = 346$ )
med =  $+0.70$  MWU  $p = 7.36e-01$ 
[hit_ramp] early ( $n = 174$ ) med =  $+1.84$  vs late ( $n = 346$ )
med =  $+1.94$  MWU  $p = 2.19e-01$ 
[fa_ramp] early ( $n = 174$ ) med =  $+0.69$  vs late ( $n = 346$ )
med =  $+1.78$  MWU  $p = 1.98e-04$ 
```